ELEC 221: Lab Report Guidelines

- Each individual who is doing the lab **must bring a pre-lab report** to the lab. The pre-lab reports will be **initialized by the TA .**
  Note: Lab 1 does not have a pre-lab assignment; please study the lab manual before attending.

- Each group must submit a **complete Lab Report (one PDF file per group)** to the onQ Dropbox.

Section	When to Complete	Notes
Lab report	During and after the lab	Record all experimental steps, results, and observations as a group while performing the lab, and after completing the lab, work together to prepare the analysis, discussion, and conclusions. **Only one shared lab report is required**, which should include the Experiment section with results, as well as the Post-Lab Analysis and Conclusion, representing the group's collective contribution and collaborative interpretation.

Note: The **final completed version** of the Lab Report must be submitted to the onQ Dropbox **by the deadline announced on onQ**.

Recommendation: We strongly encourage **all group members** to read the entire lab manual **before coming to the lab**. Having an overview of the procedure in advance will help your group follow the steps more smoothly and understand the purpose behind each part of the experiment.

ELEC221: Electric Circuits
Lab 1 Lab: Introduction to Electrical Lab Instrumentation

ELEC 221- Fall 2026 **Lab 1 Report**

Group Number:		
Date Performed:		
Section Number:		
	Group Members Name	**Group Members Student Number**

Please note that there is no pre-lab for Lab 1. Kindly review the lab manual before attending.

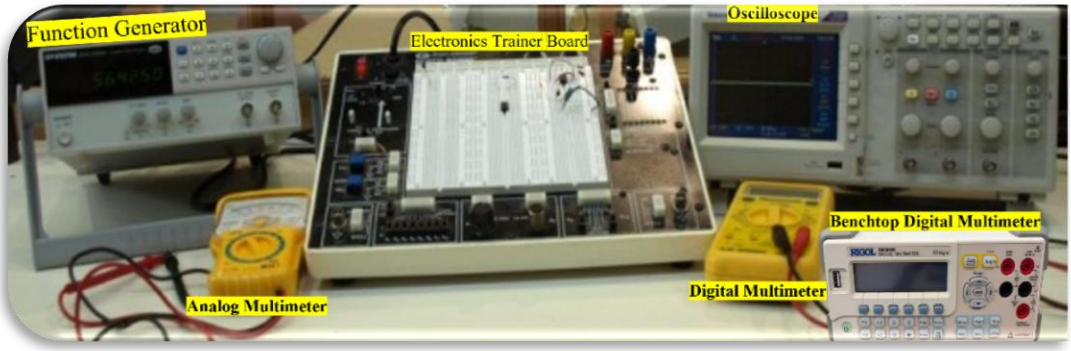
Objectives:

1. Become familiar with laboratory instrumentation and devices (proto-board, analog and digital multimeters, function generator, and oscilloscope).
2. Learn the proto-board layout: power binding posts, power rails, and how nodes are formed on a solderless breadboard.
3. Read resistor color codes and verify the values with both analog and digital multimeters.
4. Measure DC voltage (meter in parallel) and DC current (meter in series) and compare the readings with simple circuit calculations.
5. Operate the function generator to produce sine, square, and triangle waveforms at a chosen frequency and amplitude.
6. Set up the oscilloscope (probe attenuation, probe compensation, AUTOSET, and Measure menu) and correctly interpret the displayed waveform.

Definitions:

DC: Direct Current is the electricity that flows in one constant direction.
AC: Alternating Current is the electricity that periodically reverses direction.
Ohm's Law ($V = IR$): voltage equals current times resistance.
BNC port: Bayonet Neill–Concelman is a coaxial connector that locks with a quarter-turn.
TTL: Transistor-Transistor Logic is a bipolar digital family using 0–5 V levels.
CMOS: Complementary Metal-Oxide-Semiconductor is a low-power MOSFET logic family.

An overview of your toolbox for learning:



Welcome to Lab 1. In this lab we become familiar with the essential instruments used in electronics experiments. First is the Powered Breadboard Workstation, where circuits are built and powered and bread-boards. Next, are resistors, identified by color bands and then checked with a meter. Then come the multimeters, used to measure resistance, voltage, and current. After that we use the function generator to produce sine, square, and triangle signals. Finally we use the oscilloscope to see how those signals change with time. These five tools are the basic toolbox for the rest of ELEC 221.

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1. Powered Breadboard Workstation

An Electronics Trainer Board (also commonly called a Powered Breadboard Workstation, Electronics Design Trainer, or Proto-Board Station) is an all-in-one laboratory platform used for circuit prototyping, testing, and educational experiments.

The Powered Breadboard Workstation used in ELEC 221 is an analog and digital electronics design station (Fig. 1). It contains several modules. In this lab we introduce only the modules and ports used in the ELEC 221 labs; these are labeled with letters (a to h) in Fig. 1.

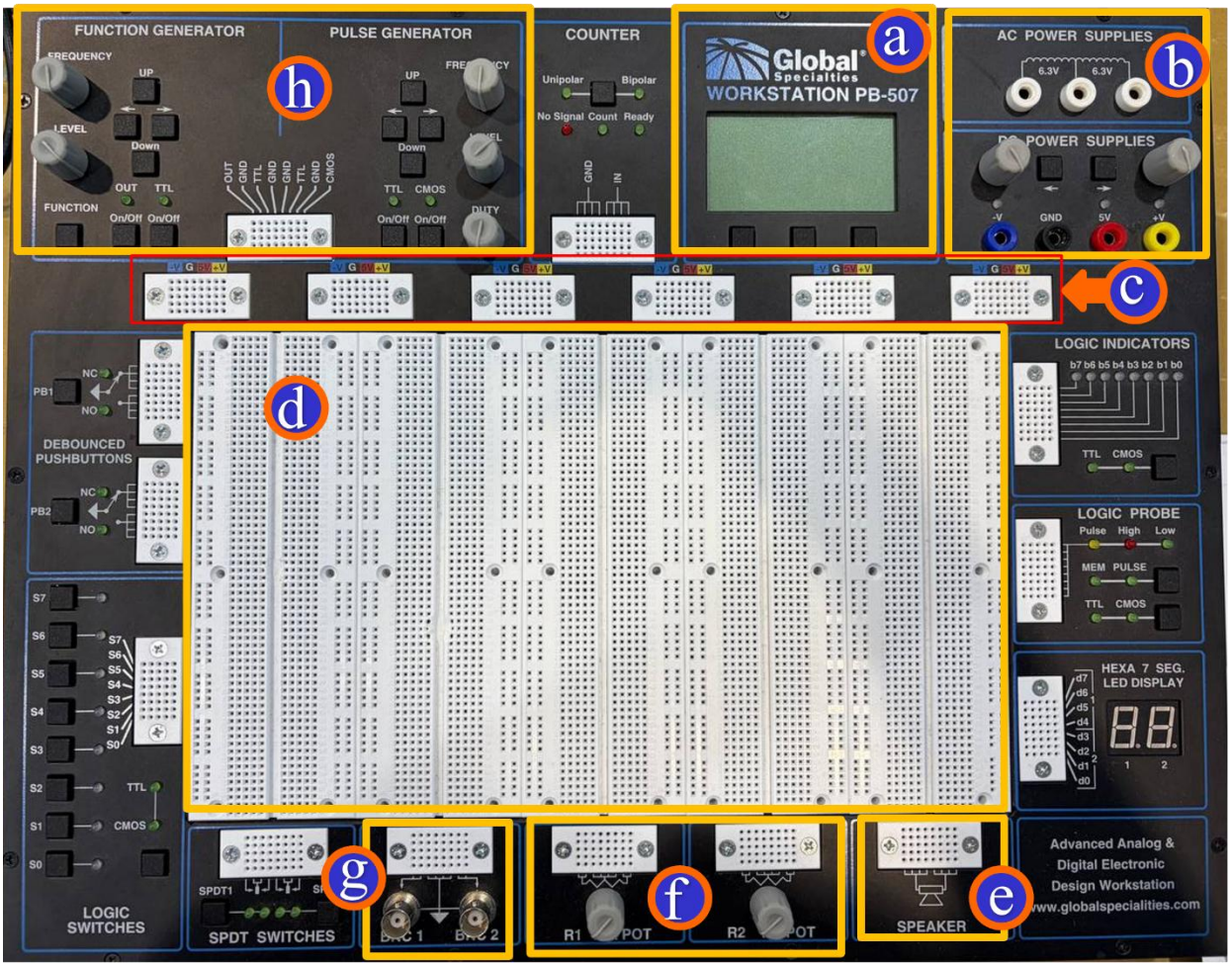
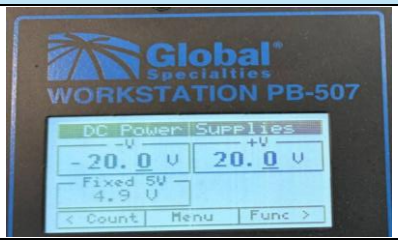


Figure 1. PB-507 Powered Breadboard Workstation front layout



Figure 2. PB-507 Powered Breadboard Workstation back layout

To power the workstation, set the ON/OFF switch on the rear panel to ON, as shown in Fig. 2. The board is ready for use when the power indicator is lit.

Powered Breadboard Workstation According to Fig. 1:	
Module Number	Description
a	The workstation has an LCD that shows voltage, current, and frequency. The screen always follows the active module: when a control on any module is used, the display switches to that module and shows its name at the top, together with its present settings. 
b , c	**The b module consist of AC power supplies and DC power supplies :** The AC Power Supplies module provides 6.3 V AC or 12.6 V AC, center-tapped, at 100 mA. The DC Power Supply module provides, a fixed +5 V DC (yellow and black banana-jack terminals) output with 1 A, a variable positive output, 0 to +20 V DC (red and blue banana-jack terminal) with 0.5 A, and a variable negative output , 0 to −20 V DC with 0.5 A. The variable positive and negative outputs are set with the adjustment knobs. Power can be taken from the banana-jack terminals built in module b, or from the tie-point sockets above each breadboard bus strip (Module c). Five of these tie-point sockets are provided. From left to right, the fixed +5 V DC, variable positive, ground, and variable negative connections are marked yellow, red, black, and blue. * **Black Post**: Serves as the **ground (GND)** connection, completing the circuit and providing a reference point for voltage measurements.
d	The breadboard has 4150 contact points (tie points). Breadboard is fully described in Section 2.
e	A built-in speaker (0.25W, 8Ω) for analog output. The connection is tie-point style.
f	A potentiometer (POT) is a variable resistor with 3 pins. Turning its knob moves an internal wiper and changes the resistance. The workstation has two built-in potentiometers:**10 kΩ POT** (left): adjustable from about 0 to 10 kΩ and **1 kΩ POT** (right): adjustable from about 0 to 1 kΩ.
g	Two BNC connectors that can connect the protoboard to external equipment (like oscilloscopes or external signal generators).
h	The workstation includes a 1 MHz function generator with sine, triangle, and square outputs (also TTL and CMOS). A pulse generator is also present; it is not used in ELEC 221, but it can serve as a second function generator. The function generator is described separately in Section 3.

2. Bread-Boards

The breadboard is a fundamental tool for prototyping and testing electronic circuits without the need for soldering. It consists of a grid of interconnected holes arranged both vertically and horizontally. The central area is typically used for placing components and making connections, while the side rails—often referred to as power buses—are designated for supplying power and grounding.

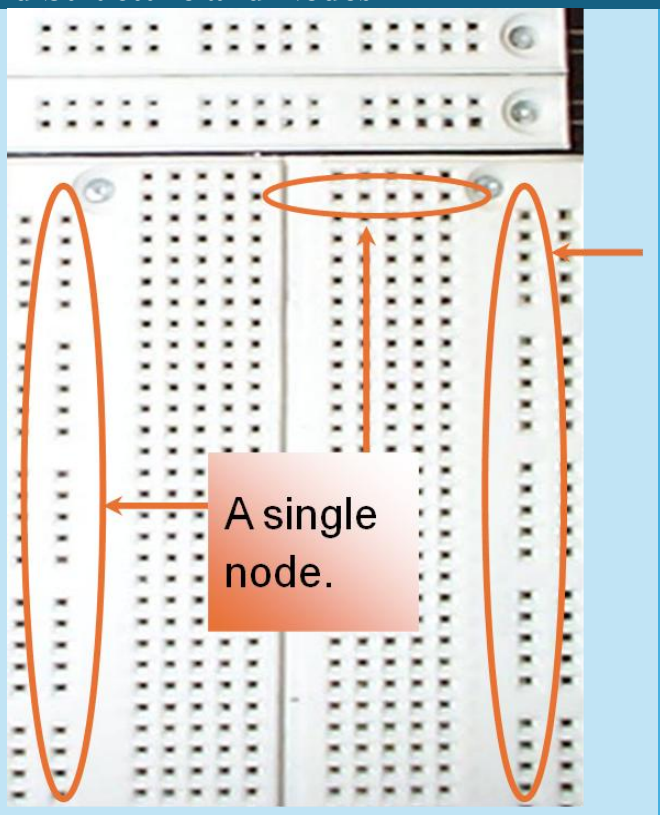
Understanding the internal structure of a breadboard is essential for effective circuit prototyping. The breadboard is designed with a grid of holes that are electrically connected in specific patterns to facilitate easy component placement and wiring.

2.1. Breadboard Structure and Nodes

1. **Horizontal Rows (Central Area):** Each row consists of 5 holes that are electrically connected, forming a single node. This means that any components inserted into the same row are connected, allowing current to flow between them.

2. Vertical Columns (Power Rails): On the sides of the breadboard, there are columns of 25 holes, typically broken into 5 groups of 5 holes. These columns are also internally connected and are commonly used for power distribution such as connecting to a constant or variable voltage source or ground.

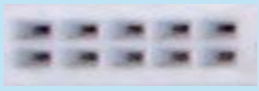
3. Node Definition: A node in an electrical circuit is a point where two or more components are electrically connected. Nodes are critical for analyzing circuit behavior, as they represent junctions where current can split or combine.



2.2. Internal Structure of a Solderless Breadboard

A solderless breadboard works because of metal (usually copper-Fig. 4) strips under the plastic. These strips line up with the holes on top and connect selected rows and columns, so components can be joined without soldering.

Horizontal and Vertical Strips: These strips are arranged to connect groups of holes into nodes, which are shared electrical points in a circuit. For example, horizontal strips connect rows of 5 holes in the central area, while vertical strips connect columns of holes in the power rails.



Node Formation: The combination of these metal strips and the hole layout creates nodes, which are defined as points where two or more circuit components are electrically connected. Nodes are essential for circuit operation, as they allow current to flow between components and enable signal transmission.



Figure 4. Breadboard's underneath

2.3. Connecting Components on a Breadboard

To build circuits on a solderless breadboard, components are connected by inserting their metal leads (or "legs") into the holes on the board. These holes are internally connected by metal strips, forming nodes that allow electrical signals to pass between components.

Shared Nodes: Components that need to be electrically connected should be placed into holes that belong to the same node. In the central area of the breadboard, this means using the same row of 5 connected holes. On the side power rails, components can be connected using the same column of 25 holes, which are typically used for power and ground distribution.

Efficient Prototyping: This method of connection allows for quick and flexible circuit assembly without soldering. It also makes it easy to modify or troubleshoot circuits by simply repositioning components or wires.
Fig. 5 shows, how components are inserted and connected using shared nodes, providing a visual reference for proper breadboard usage.

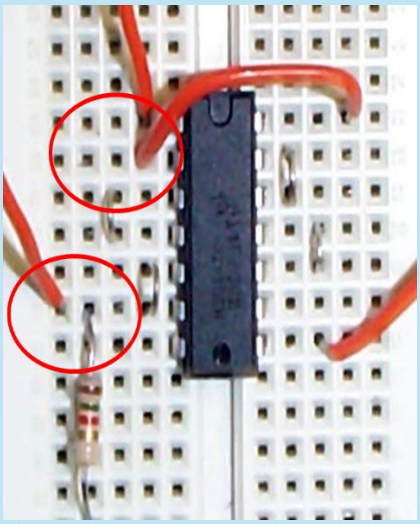


Figure 5. how components are inserted and connected using shared nodes

3. Resistor Color Codes

This Fig. 6, illustrates the resistor color code system, which is used to determine the resistance value of a resistor based on the colored bands printed on its body. Each resistor typically has four color bands, and each band represents a specific part of the resistance value. This coding system allows for quick identification of resistor values without needing to measure them. Each lab bench is equipped with resistors like the one shown, enabling students to practice reading color codes and selecting appropriate resistors for their circuits.

First Band: Indicates the first digit of the resistance value.
Second Band: Represents the second digit of the resistance value.
Third Band: Acts as the multiplier, determining how many zeros to add to the base value.
Fourth Band: Specifies the tolerance, which tells how much the actual resistance may vary from the stated value (e.g., ±5%).

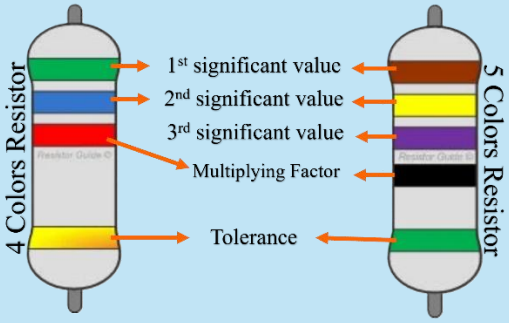


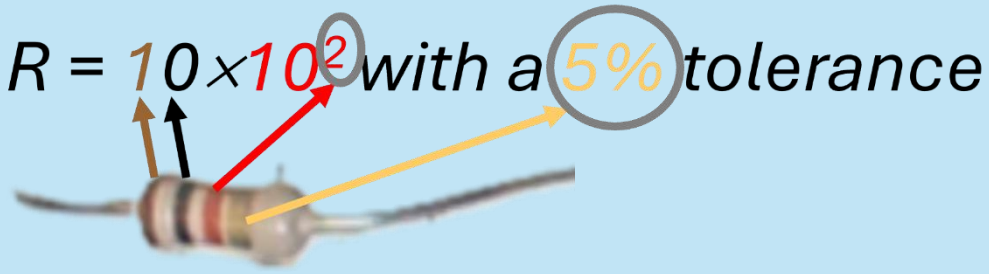
Figure. 6. Resistor 4 and 5 Color Codes

3.1. Resistor Color Code Table

The below table presents the resistor color code, which is used to determine the resistance value and tolerance of resistors based on the colored bands printed on their bodies. Each color corresponds to a specific digit, multiplier, or tolerance level. For example, Red represents digit 2 and a tolerance of 2%, while Gold and Silver are used for lower tolerance values. This table is a quick reference for identifying resistor values during circuit design and testing.

Colour		Digit	Multiplier	Tolerance
Black		0	10^0	
Brown		1	10^1	
Red		2	10^2	2%
Orange		3	10^3	
Yellow		4	10^4	
Green		5	10^5	
Blue		6	10^6	
Violet		7	10^7	
Grey		8	10^8	
White		9	10^9	
Silver			10^{-2}	10%
Gold			10^{-1}	5%
None				20%

With this example we are going to ddemonstrate how to calculate the resistance value of a resistor using its color bands. This means the resistor has a value of 1000 ohms (1 kΩ) with a tolerance of ±5%, indicating the actual resistance may vary between 950 Ω and 1050 Ω :



Important Note: Measurement tools vs Stimulus tools

The core distinction you made is excellent: multimeters and oscilloscopes are measurement (output) tools, while power supplies and function generators are stimulus (input)

In an electronics laboratory, equipment is divided into two primary categories: measurement devices and stimulus sources. Devices like analog multimeters, digital multimeters, and oscilloscopes are measurement tools (Section. 4). They capture and analyze a circuit's output signals without altering them. Conversely, power supplies (already discussed) and function generators (Section. 5) act as input sources that inject electrical energy into a circuit to make it operate. While a standard DC power supply provides a steady, constant voltage to power the circuit, a function generator provides controlled, time-varying electrical signals (waveforms).

4. Measurement: The First Skill in the Lab

Accurate measurement is a cornerstone of electrical engineering. In any circuit, whether complex or straightforward, understanding the behavior of electrical quantities such as voltage, current, and resistance is essential for design, analysis, troubleshooting, and optimization. This section introduces the essential measurement tools used to analyze circuit performance in real time: analog multimeters, digital multimeters, and oscilloscopes.

Without precise measurements, circuits cannot be reliably tested or validated. For example, verifying voltage levels ensures components operate within safe limits, while current measurements help prevent overloads and identify inefficiencies. In this section we will introduce

An **analog multimeter** measures voltage, current, and resistance using a physical, moving needle over a printed scale. It continuously sweeps back and forth, making it excellent for tracking real-time fluctuations or gradual trends in a signal, though it requires you to manually read the needle alignment.

A **digital multimeter (DMM)** measures the same electrical properties but converts the data into a discrete numerical value displayed on a screen. DMMs are highly accurate, automatically select measurement ranges, and provide exact numbers, eliminating human reading errors.

An **oscilloscope** goes beyond simple numbers to graph voltage over time on a screen. Instead of a static value, it visualizes the entire electrical waveform, allowing you to measure properties of AC and high-frequency signals.

4.1. Measurement with Analog Multimeter

An analog multimeter is a key instrument used to measure current, voltage, and resistance in electrical circuits. The multimeter features a rotary knob that must be set appropriately depending on the measurement type.

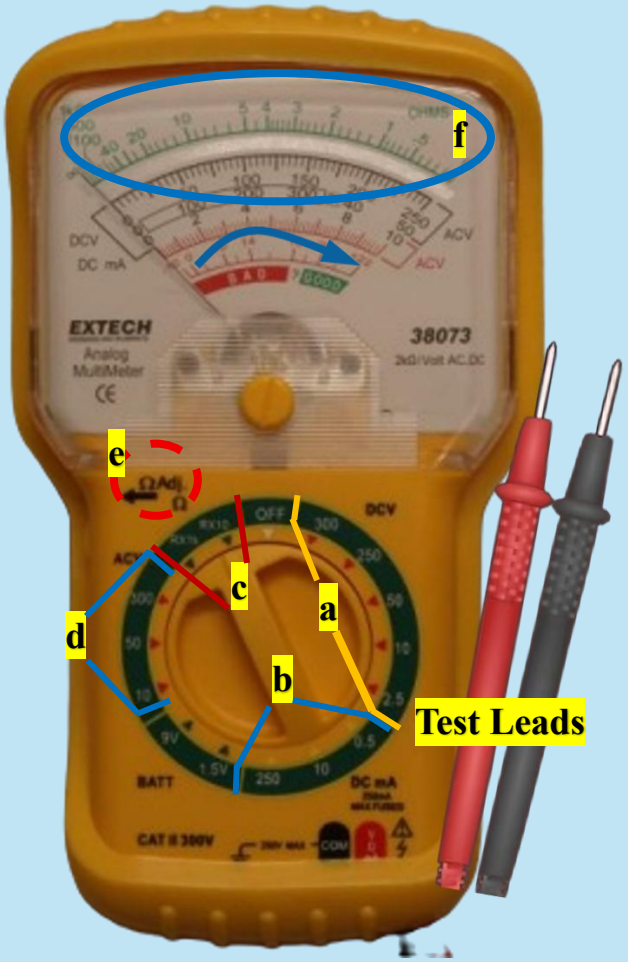


Figure.7. Analog Multimeter

a. Measure DC voltage by turning the knob to DCV

b. Measure the current by turning the knob to DC mA

c. Measure the resistance by turning the knob to RX10Ω or RX1kΩ (for low or higher resistance)

d. Turn the knob to ACV to measure AC voltage

e. The knob located on the left is used to zero the meter when measuring resistance

Activity 1: Short Techniques

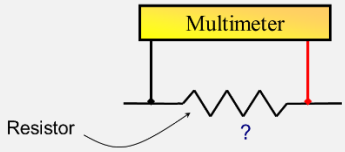
Connect (short) the leads of the two probes (test leads) together. The meter needle (part f in Fig. 7) should go to zero ohms on the resistance readings scale (the very top scale that is in green). Use the adjusting knob to make it read zero.

Activity 2: Measurement Techniques-Resistance

In this activity, each group will practice measuring resistance values using both **analog and digital multimeters** and compare the results with the expected values derived from resistor color codes.

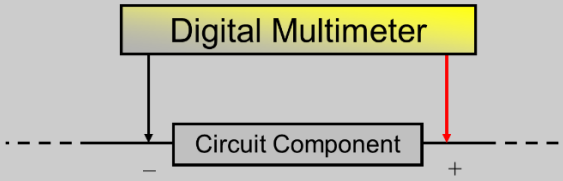
1. **Identify the Resistors**: Locate the **1kΩ, 510Ω, and 680Ω** resistors on your lab bench. *Hint: Use the resistor color code table to calculate the theoretical resistance value.
2. **Measure with Multimeters**: Use each multimeter (analog and digital) to measure the resistance of each resistor.

3. **Compare Values:** Check if the measured values fall within the tolerance range indicated by the fourth color band.



Activity 3: Measurement Techniques - Voltage

In this activity we will parallelly measure the voltage.
When a multimeter is used to measure voltage, it operates as a voltmeter. To accurately measure the voltage across a circuit component, the voltmeter must be connected in parallel with that component. This ensures that the voltage drop across the component is directly displayed on the multimeter without altering the circuit's behavior.



This technique is essential for verifying power supply levels, checking voltage drops across components, and diagnosing circuit issues.

How to measure the voltage:
1**. Set the Multimeter:** Turn the knob to DCV to measure DC voltage.
2. **Connect the Probes:** Place the positive (red) probe on the positive side of the component. Place the negative (black) probe on the negative side or ground.
3. **Read the Display:** Observe the voltage reading on the multimeter screen.
Ensure the multimeter range is appropriate for the expected voltage level.
Activity Instructions: follow Fig. 8.
1. Set the digital multimeter to DC voltage (DCV).
2. Connect the red probe to the positive variable-voltage binding post on the DC power supplies module of the Powered Breadboard Workstation. Connect the black probe to the ground binding post.
3. Turn the positive variable-voltage (yellow) knob and watch the meter reading.
4. Adjust the knob until the meter shows approximately +10 V.



Figure.8. The connections for this activity

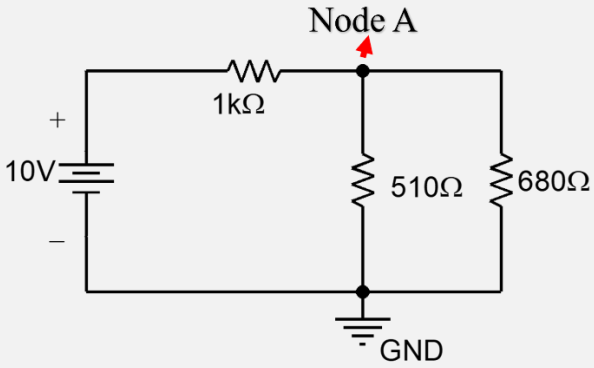
Activity 5: Build a circuit

In this exercise, each group will construct a circuit on the **breadboard** of the Powered Breadboard Workstation and use a digital multimeter to measure voltage across different components. This activity reinforces the concept of measuring voltage by connecting the multimeter in parallel with the element of interest.

Tasks:

Use a 10V DC power source (the positive variable-voltage (yellow) and ground (black), adjust the designated knob to see the 10V).

Use the resistors and jumper wires that are provided for you and build this circuit below:



1. From Connect a 1kΩ resistor from the positive terminal to Node A.
2. From Node A, connect a 510Ω resistor and a 680Ω resistor in series to ground (GND).

Fig. 9 displays the under-test circuit:

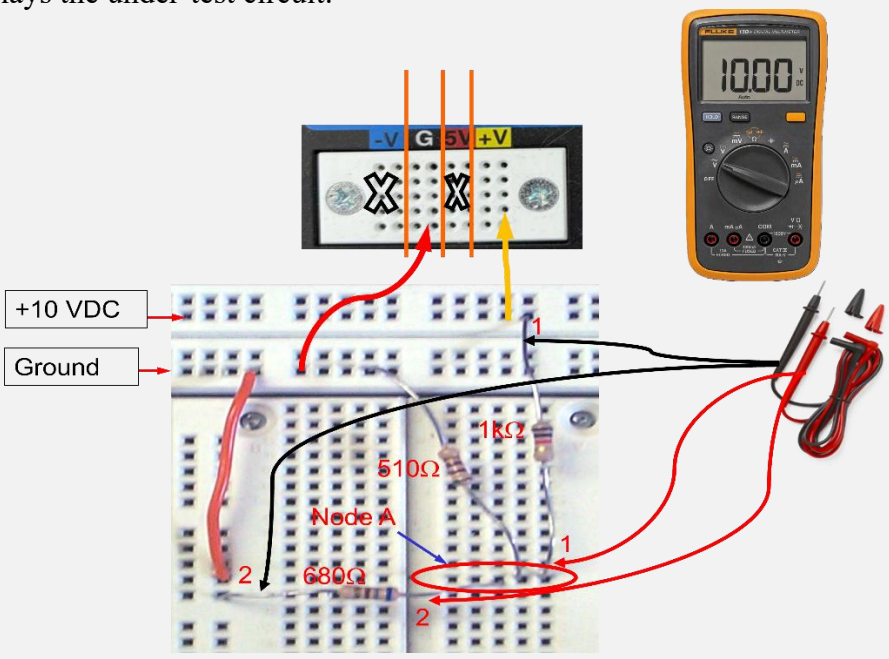


Figure. 9. The test circuit

Now Measure Voltages, Record the values and compare them with theoretical expectations based on voltage division:

Resistors	Theoretical voltage (V)	Measured voltage (V)
1 kΩ		
510 Ω		
680 Ω		

Conclusion: When using measurement instruments in a circuit, it's essential to understand that they can influence the behavior of the circuit itself (acting like a resistor). This is especially true for voltmeters and multimeters.

In the ELEC 221 lab experiments, the digital multimeters used have sufficiently high impedance, making their impact on circuit measurements negligible in most cases. However, it is still essential to be aware that:

1. The measurement instrument becomes part of the circuit when connected (acting like a resistor).
2. In sensitive or high-impedance circuits, even small currents drawn by the meter can affect reading.
3. Proper connection techniques, such as placing the voltmeter in parallel with the component being measured, are essential to minimize interference.

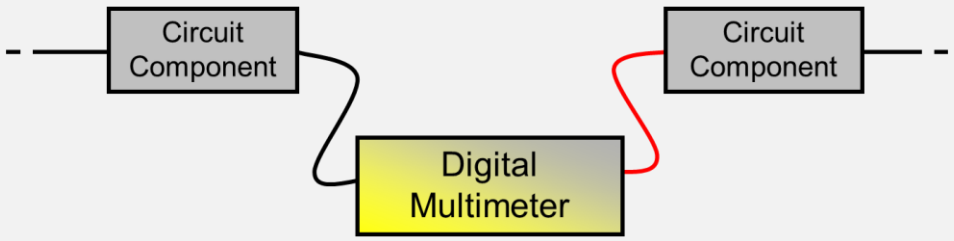
Understanding these effects helps you accurately interpret measurements and recognize when instrument limitations might influence their results.

Activity 6: Measurement Techniques-Current

When a multimeter is used to measure current, it operates as an **ammeter**. Unlike voltage measurement, where the meter is connected in parallel, current measurement **requires the meter to be connected in series with the circuit component**. This ensures that the current flowing through the component also flows through the meter.

How to measure current:

1. **Set the Multimeter:** Turn the multimeter knob to the appropriate current (A) setting
2. **Break the Circuit:** Identify the location in the circuit where you want to measure current. Then open the circuit at that point so the multimeter can be inserted.
3. **Connect in Series:** Connect one probe of the multimeter to one side of the break. Connect the other probe to the other side, so the current flows through the meter.
4. **Read the Display:** Observe the current reading on the multimeter.



Activity Instructions: Follow Fig. 10.

In this exercise, each group will measure the current flowing through a circuit component using a multimeter configured as an ammeter. This activity reinforces the concept of series connection for current measurement.

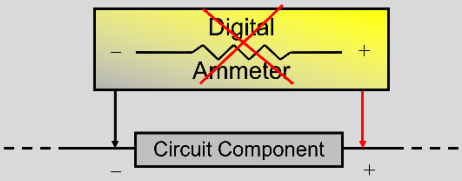
Use the circuit already built (10 V DC, 1 kΩ, 510 Ω, and 680 Ω).

1. **Measure the current for** 680 Ω **only.**
2. **Set the multimeter to a suitable current (A) range. Plug the probes into the current terminals.**
3. **Open the circuit at that point and connect the meter in series across the break so the same current flows through the meter.**
4. **Read the current and record it.**

Resistor	Measured current (mA)	Theoretical current using Ohm's Law (mA)
680 Ω		

Warning: Never connect the meter across a voltage source while it is set to current. That can blow the fuse.

WARNING: a common way to blow a multimeter fuse is to measure voltage while the meter is still set to current (A). In current mode the meter is an ammeter and has very low resistance. If it is placed across a voltage source, a large current flows and the fuse opens. Check the setting before you connect the probes: Use V for voltage (parallel) and A for current (series) and never put an ammeter across a supply.



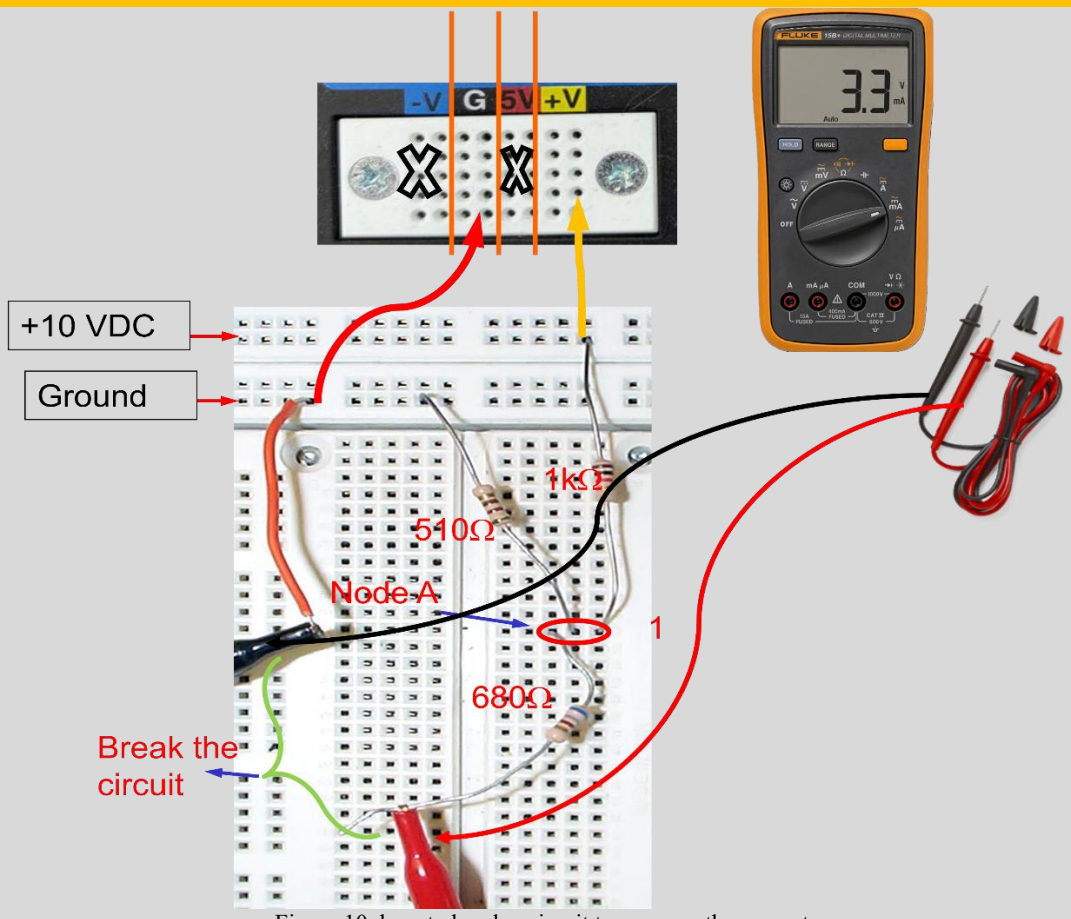


Figure.10. how to break a circuit to measure the current

Conclusion: Ideally, an ammeter should have no resistance, so it does not affect the circuit it is measuring. In reality, an ammeter does have some resistance, and this can influence the circuit slightly. The **measurement instrument** becomes a part of the circuit, and this effect should be kept in mind. In the ELEC 221 lab, the digital multimeters used have very **low internal resistance**, so their impact on the circuit is usually **negligible**.

4.2. **Oscilloscope (** Tektronix TDS2001C**)**

An **oscilloscope** is essentially a test instrument used to visualize electrical signals. Instead of just giving you a single number like a standard multimeter, it draws a real-time graph where the vertical axis (Y) represents voltage, and the horizontal axis (X) represents time. This allows engineers and technicians to see exactly how electrical signals behave, change, or experience distortions over time.

The Tektronix TDS2001C is a specific 2-channel Digital Storage Oscilloscope (DSO).

In this section we will learn how to use the oscilloscope through activities, Fig. 11. Shows the schematic of oscilloscope:

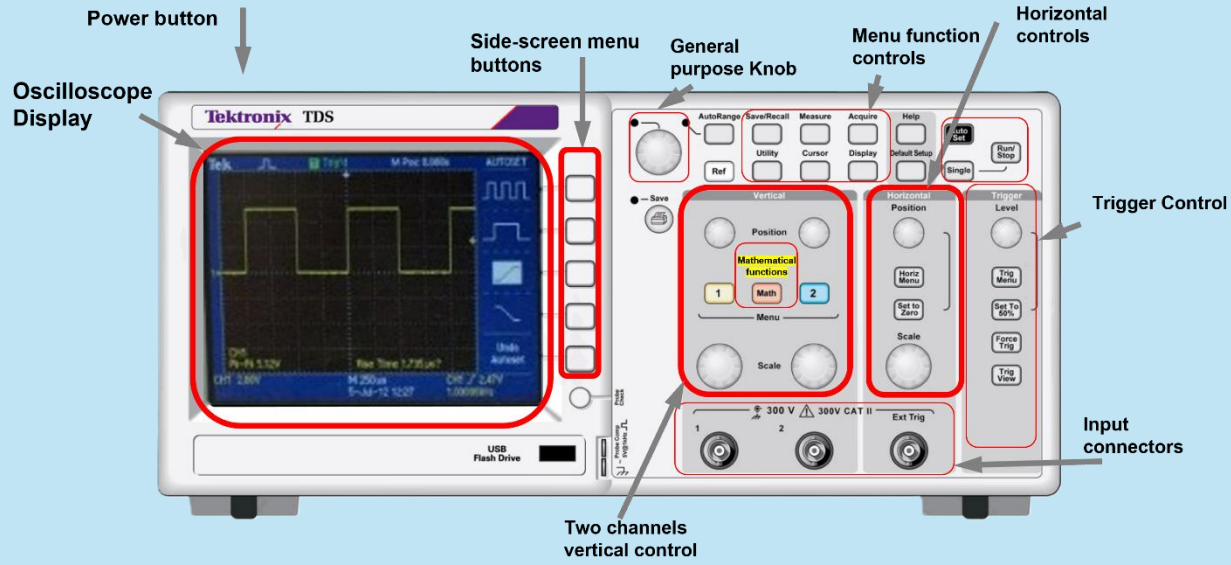
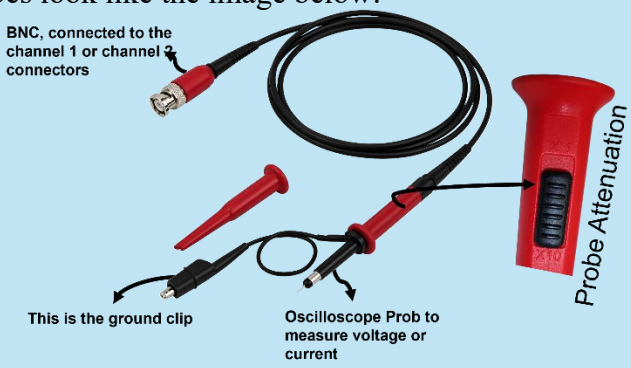


Figure 11. Schematic of **Oscilloscope (**Tektronix TDS2001C**)**

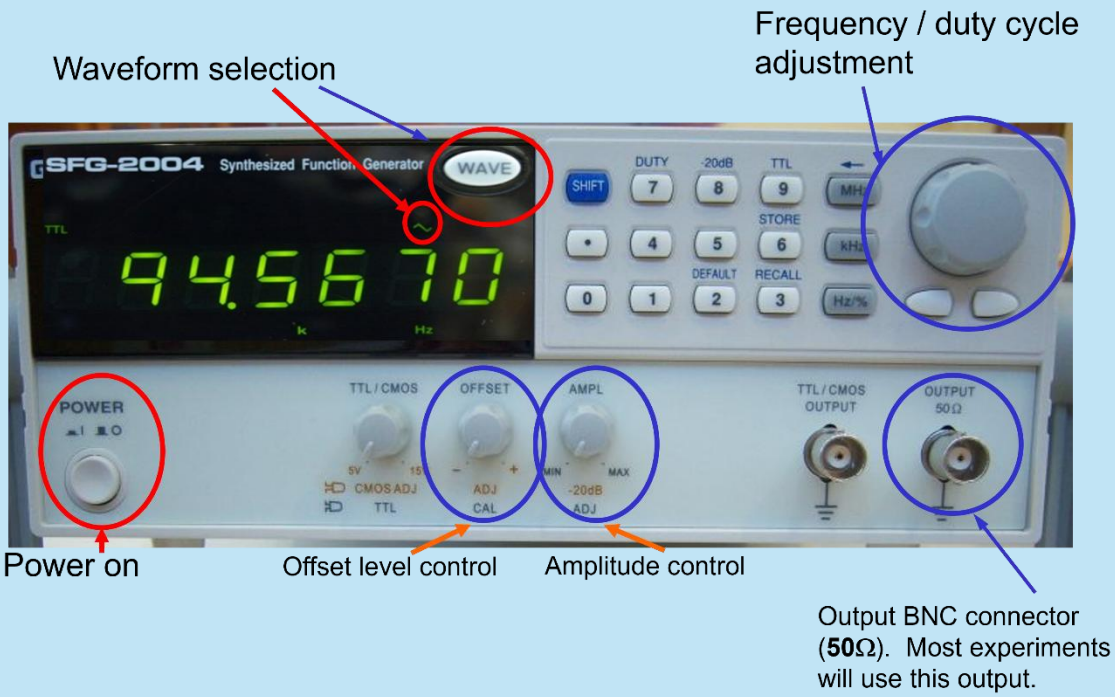
The Tektronix TDS2001C has two independent channels, so two nodes in a circuit can be observed at the same time. Each probe carries the voltage/current/frequency at its node to the oscilloscope. The probes look like the image below:



Probe attenuation settings is important when using an oscilloscope. If your probe includes an attenuation switch, it should be set to x10 for most measurements. **This setting increases the impedance of the probe**, which helps to reduce loading effects on the circuit under test. Lower loading ensures more accurate signal readings and minimizes interference with the circuit's behavior. The typical probe shown above, with a visible attenuation switch and includes a reference.

5. Function Generator

A **function generator** is an electronic test instrument used to inject controlled, predictable electrical signals into an electrical circuit. In a circuit, the function generator acts as the input stimulus.



When the function generator drives a circuit, three quantities are set:

1. **Waveform:** how the voltage changes with time

Sine: smooth wave ،Square: sudden high/low steps and Triangle: straight-line rise and fall

2. **Frequency (Hz):** how many times the wave repeats each second. Example: 1 kHz = 1000 cycles per second.
3. **Amplitude (V_{pp}):** the height of the wave from the lowest point to the highest point (peak-to-peak voltage).

Amplitude knob: Turning the knob changes only the height of the wave, not the frequency. Clockwise = larger V_{pp}; counterclockwise = smaller V_{pp}. On many generators, pulling the knob out applies a −20 dB attenuator and divides the amplitude by 10 (for example 10 V_{pp} becomes 1 V_{pp}). Use this when a small, precise signal is needed.

DC offset knob: By default, the wave is centered on 0 V (half positive, half negative). Offset adds a constant DC level and moves the whole wave up or down without changing its height. On many generators the offset is off while the knob is pushed in. Pull the knob out to enable offset, then turn it to shift the waveform. Example: pull out and raise the wave until its bottom sits at 0 V if a 0-to-5 V signal is required.

Activity 7: Function Generator Test Drive

In this exercise, each group will connect a function generator to a speaker (built in the Powered Breadboard Workstation) to observe how electrical signals can be converted into sound. This hands-on test helps students understand signal generation and output behavior.

This test drive helps you to visualize and hear the effect of waveform generation and understand how electronic signals can be used in real-world applications like audio systems.

Prepare the Equipment: Use a coaxial cable with alligator clips (Fig. 12) attached to one end. Locate the 50Ω output on the function generator.

Connect the Function Generator:

Plug the coaxial cable into **the BNC output** jack (j2) of the function generator.

Attach the red and black alligator clips to the 1 and 2 terminals of the 8Ω speaker (built in the Powered Breadboard Workstation).

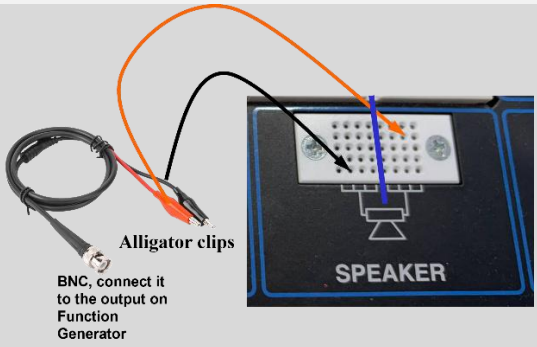


Figure. 12. A coaxial cable with alligator clips

Now Turn on the function generator. Select a waveform (e.g., sine wave) and set a frequency that is audible (e.g., 1 kHz). Adjust the amplitude to a safe level for the speaker.

Observe the Output:

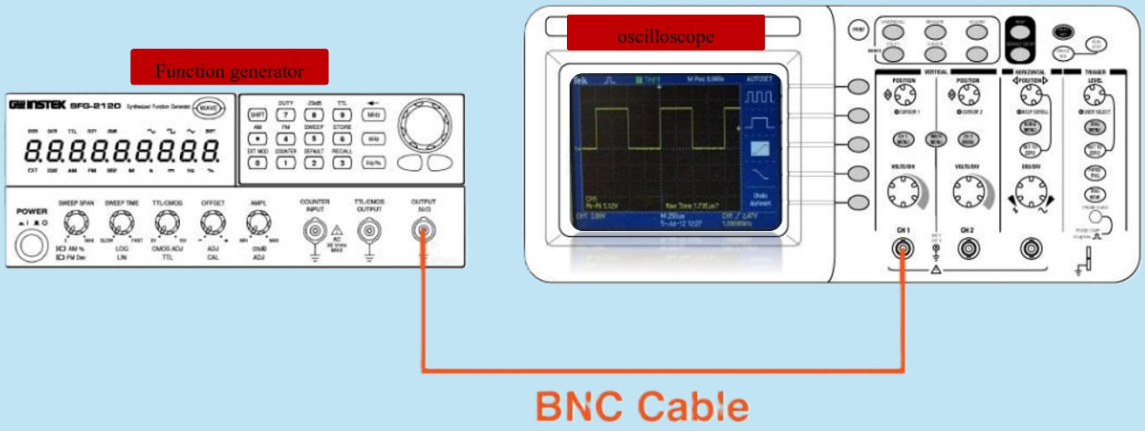
Listen to the sound produced by the speaker.

Try changing the waveform and frequency to hear how the output changes.

Section 6. Co-operation of Function Generator and Oscilloscope

The combination of a Function Generator and an Oscilloscope forms the ultimate "duo" in electronics engineering. They are almost always used together because they represent the two halves of circuit testing: Input (Stimulus) and Output (Measurement).

Think of the **function generator as the Question you ask a circuit**, and **the oscilloscope as the circuit's Answer.**



Activity 8: combination of a Function Generator and an Oscilloscope

In this exercise, each group will learn how to operate a function generator and use it to provide input signals to electronic circuits. The function generator is a versatile instrument used to produce various types of waveforms such as sine, square, and triangle waves at different frequencies and amplitudes.

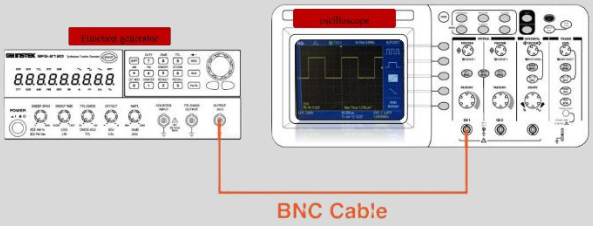
Activity instrumentation:

1. Power On the Device

2. Set the Output Parameters: Use the frequency knob to adjust the desired signal frequency. Select the waveform type (sine, square, triangle) using the appropriate function selector. Adjust the amplitude and DC offset if needed.

3. Connect to the Oscilloscope: Using BNC cables or probes leads to connecting the output of the function generator to the input of your Oscilloscope /circuit (Ch1 or Ch2).

4. Use an oscilloscope to observe the waveform and confirm they match the settings on the function generator.



Activity 9: Testing the Probe

1. In oscilloscope connect the CH1 probe to the compensation tab (Fig.13).

2. Initial Setup: Wait for the oscilloscope to boot up and display the default screen.

Familiarize yourself with the layout: the screen is on the left, and the control buttons and knobs are on the right.

3. Prepare for Measurement: Connect the probe to Channel 1. Attach the probe tip and ground clip to the test points on your circuit. Adjust the vertical and horizontal scales to properly display the waveform.

4. Observe and Analyze: Use the AUTOSET button to automatically configure the display for a clear view of the signal.

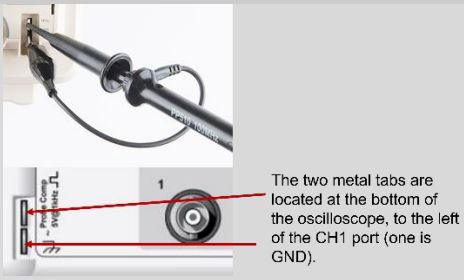
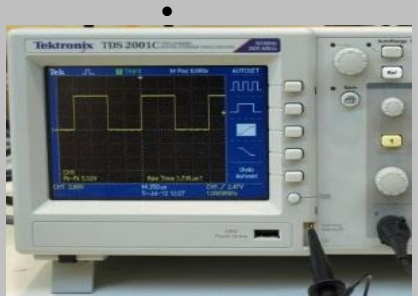


Figure. 13. Probe comp

2. You should get this:



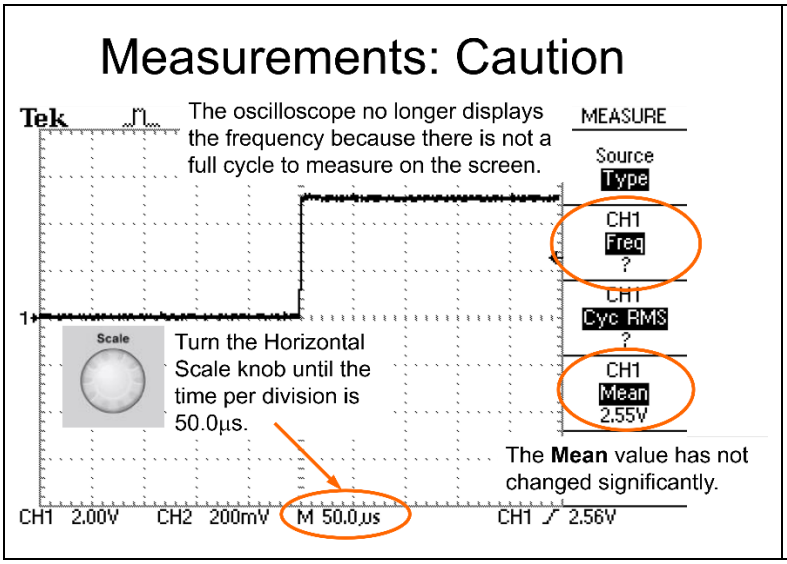
Now adjust the probe so that the top of the square wave is as flat as possible.

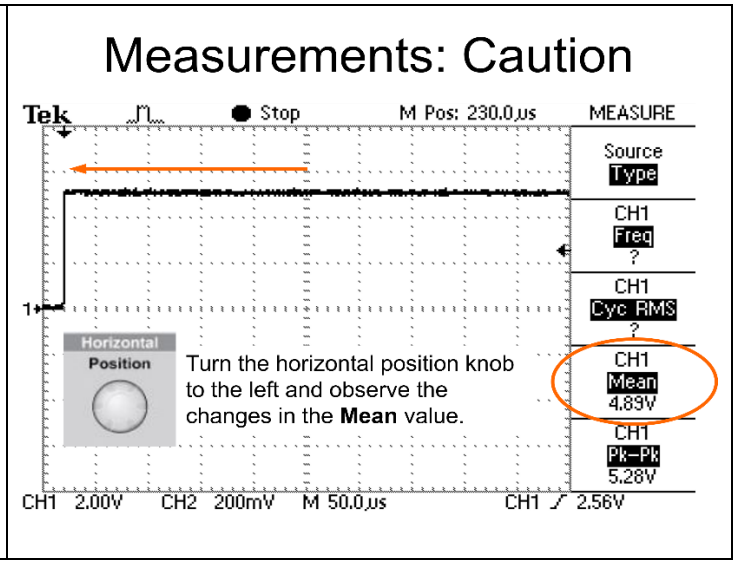


Figure. 13 Scope side bottoms

Follow Fig. 13 and press the Measure button and this menu will appear, you can also Press a button beside a measurement to select the source and type. Also, you can select the source and type of measurement, then press the “back” button.

Following Fig 14 and 15 and the Mean value increases as the waveform is shifted to the left, gradually approaching the peak-to-peak voltage. The mean voltage of the signal has not really changed. However, the mean value of the waveform that is displayed on the screen has changed. The measurement functions of this oscilloscope operate on the displayed waveform, not on the actual input signal, and must be interpreted with care.





Lab 1 – Group Knowledge Check & Report Submission
Instructions for Student Groups

As part of your Lab 1 submission for ELEC 221, your group is required to collectively respond to the following set of questions as part of your final Lab Report PDF Upload the completed report to the OnQ Dropbox under your group number, as outlined in the lab guidelines. These questions are designed to reinforce your understanding of the lab equipment, measurement techniques, and theoretical concepts covered in the lab manual and activities.

Lab 1 – Questions for Report Preparation

Resistor Color Code & Identification:

1. What is the resistance value of a resistor with the color bands: Brown, Black, Red, Gold?
2. Which color bands would represent a 680Ω resistor with a ±5% tolerance?
3. If a resistor is measured to be 510Ω, what color bands should you expect to see?
4. What does the fourth color band on a resistor indicate?
5. How does the multiplier band affect the total resistance value?

Multimeter Usage & Measurement Techniques

7.What setting should the analog multimeter knob be when measuring DC voltage?
8. How can an analog multimeter be calibrated?
9.What is the correct multimeter setting to measure AC voltage?
10.When measuring current, how should the multimeter be connected in the circuit?
11.What is the most common cause of a blown fuse in a multimeter?

Breadboard & Circuit Construction

12. What is the purpose of the power rails on the sides of the breadboard?
13. What is a node in an electrical circuit?
14. Does setting the probe attenuation to x10 help reduce?
15. What waveform is generated by the PROBE COMP tab on the oscilloscope?
16. What does the AUTOSET button on the oscilloscope do?
17. How can you measure that voltage across the 680Ω resistor?